ENSAI École nationale de la statistique et de l'analyse de l'information

TP 3 Supervised Learning 2A
2025-2026

Use of ChatGPT / Claude / other tools encouraged.
The goal of this practical session is to work with CART trees and random forests.

Part 1: Loading the dataset, exploratory analysis

The dataset includes measurements taken on wine, as in TP 2. For each wine sample, 11 characteristics were measured (acidity, density, pH, etc.), and after tasting, each received a score. Unlike TP 2, however, you are given the actual score (and not the good quality / bad quality class), so this is a regression problem.

Questions:

1. Load the dataset. Perform an exploratory analysis to highlight the characteristics of the data.

2. Using a preliminary visual exploration of the data, identify the features that seem at first glance to have the most influence on wine quality.

Part 2: Sample splitting and model training

In this part, you will train CART tree models and random forests. You will use the ranger package for random forests.

1. Build the necessary datasets for supervised learning.

2. CART Algorithm
    a. Train a CART tree.

    b. Identify in the result the competing and surrogate splits for the root node. For this node, what split would the tree use to predict a new wine for which the alcohol content and density were not measured and are therefore missing?
    (Remember to review this section in the course notes!)

    c. Visualize the CART tree. Using just the diagram, what would be the prediction for a new wine with the following characteristics: alcohol = 11.5, pH = 2, free.sulfur.dioxide = 7.6, sulphates = 0.42, volatile.acidity = 0.98?

    d. Compute the generalization error of this tree and propose a visualization to compare true values with predicted ones.

    e. Now, let's look at the robustness of CART trees. By generating several new training samples (each with a size equal to 50% of the original data), illustrate that the CART tree is not robust. Comment.

3. Random Forests
    a. Using the documentation of the caret package, identify the hyperparameters of a random forest built with the ranger package that you can estimate via cross-validation.

    b. Run the cross-validation procedure by varying them and visualize the validation error during the process.

    c. Identify their optimal values and retrain a forest with these values.

    d. Compute the generalization error of the forest, visualize the differences between true values and predictions as for the CART tree, and compare the error with the Out-Of-Bag (OOB) error. **(Remember to review this section in the course notes!)**

4. Bonus: Visualization of forest predictions in a 2D projected space

    a. Install the reticulate package and the Python umap package.

    b. Normalize the training set data, then compute the UMAP projection for the normalized data. Use the supervised projection option by reading the Python umap documentation. Visualize the projection.

    c. Using the previous visualization, define a 100x100 grid covering the projected space. Compute the inverse projection of all grid points using “inverse_transform” (see Python umap documentation). After this step, remember to de-normalize the values.

    d. Compute the random forest predictions for these inverse and de-normalized points.

    e. In the 2D projected space, represent the grid points and their predictions using a heatmap; you can also overlay the projection of the training samples.